

F. Odd Queries on Odd Array

time limit per test: 10 seconds
memory limit per test: 1024 megabytes

An array b of length m is *cute* if there do not exist four indices $1 \leq i < j < k < l \leq m$ such that $b_i \neq b_j$, $b_i = b_k$ and $b_j = b_l$.

The *beauty* of an array b of length m is defined as the sum of all distinct values that appear an odd number of times in array b . Formally, let $\text{cnt}(b, x)$ denote the number of times the value x appears in array b . Then, the beauty is given by

$$\sum_{\substack{x \in \mathbb{Z} \\ \text{cnt}(b, x) \text{ is odd}}} x.$$

You are given a cute array a of length n , and you need to answer q queries online. Each query consists of two integers l and r ($1 \leq l \leq r \leq n$), and you must compute the beauty of the subarray $a_{l \dots r}$ *. Note that the queries are encoded; each subsequent query can only be decoded after calculating the answer to the preceding query.

*The subarray $a_{l \dots r}$ refers to the contiguous segment of the array a that starts at index l and ends at index r , i.e., $[a_l, a_{l+1}, \dots, a_r]$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and q ($1 \leq n, q \leq 5 \cdot 10^5$) — the length of array a and the number of queries.

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$) — the elements of the cute array a .

The i -th of the next q lines contains two integers x'_i and y'_i ($1 \leq x'_i, y'_i \leq n$) — the endpoints of the queried subarray in an encoded form.

Let ans_i be the answer to the i -th query, with $\text{ans}_0 = 0$. Then, we calculate $x_i = ((x'_i - 1 + \text{ans}_{i-1}) \bmod n) + 1$ and $y_i = ((y'_i - 1 + \text{ans}_{i-1}) \bmod n) + 1$. The endpoints of the i -th queried subarray, l_i and r_i , are decoded as $l_i = \min(x_i, y_i)$ and $r_i = \max(x_i, y_i)$.

It is guaranteed that the given array a satisfies the conditions of a cute array.

It is guaranteed that the sum of n and the sum of q over all test cases does not exceed $5 \cdot 10^5$.

Output

For each test case, output q integers representing the answers to each query.

Example

input

Copy

```
3
11 4
1 1 2 2 3 3 3 2 2 1 1
7 10
5 11
8 6
2 8
6 2
1 3 2 3 4 3
```

Codeforces Round 1057 (Div. 2)

Contest is running

01:24:53

Contestant



→ Submit?

Language: Python 3.13.2
Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	430
Problem B	645
Problem C	1075
Problem D	1505
Problem E	1935
Problem F	2795
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:35 from the first attempt

```

1 6
1 4
3 6
3 3 3
1 1
1 2
3 1
2 2
2 3
3 3

```

output

Copy

```

4 5 4 0
10 6
3 0 3 3 0 3

```

Note

In the first test case, the queries are as follows:

$$x_1 = ((7 - 1 + 0) \bmod 11) + 1 = 7, \quad y_1 = ((10 - 1 + 0) \bmod 11) + 1 = 10$$

$$l_1 = \min(7, 10) = 7, \quad r_1 = \max(7, 10) = 10$$

Thus, the queried subarray is $a_{7...10} = [3, 2, 2, 1]$. Values 3 and 1 appear once (odd), and value 2 appears twice (even). Hence, the beauty is $3 + 1 = 4$.

$$x_2 = ((5 - 1 + 4) \bmod 11) + 1 = 9, \quad y_2 = ((11 - 1 + 4) \bmod 11) + 1 = 4$$

$$l_2 = \min(9, 4) = 4, \quad r_2 = \max(9, 4) = 9$$

Thus, the queried subarray is $a_{4...9} = [2, 3, 3, 3, 2, 2]$. Values 2 and 3 appear three times (odd). Hence, the beauty is $2 + 3 = 5$.

$$x_3 = ((8 - 1 + 5) \bmod 11) + 1 = 2, \quad y_3 = ((6 - 1 + 5) \bmod 11) + 1 = 11$$

$$l_3 = \min(2, 11) = 2, \quad r_3 = \max(2, 11) = 11$$

Thus, the queried subarray is $a_{2...11} = [1, 2, 2, 3, 3, 3, 2, 2, 1, 1]$. Values 1 and 3 appear three times (odd), and value 2 appears four times (even). Hence, the beauty is $1 + 3 = 4$.

$$x_4 = ((2 - 1 + 4) \bmod 11) + 1 = 6, \quad y_4 = ((8 - 1 + 4) \bmod 11) + 1 = 1$$

$$l_4 = \min(6, 1) = 1, \quad r_4 = \max(6, 1) = 6$$

Thus, the queried subarray is $a_{1...6} = [1, 1, 2, 2, 3, 3]$. Values 1, 2, and 3 each appear twice (even). Hence, the beauty is 0.

In the second test case, the queries are as follows:

$$x_1 = ((1 - 1 + 0) \bmod 6) + 1 = 1, \quad y_1 = ((6 - 1 + 0) \bmod 6) + 1 = 6$$

$$l_1 = \min(1, 6) = 1, \quad r_1 = \max(1, 6) = 6$$

Thus, the queried subarray is $a_{1...6} = [1, 3, 2, 3, 4, 3]$. Values 1, 2, and 4 appear once (odd), and value 3 appears three times (odd). Hence, the beauty is $1 + 2 + 3 + 4 = 10$.

$$x_2 = ((1 - 1 + 10) \bmod 6) + 1 = 5, \quad y_2 = ((4 - 1 + 10) \bmod 6) + 1 = 2$$

$$l_2 = \min(5, 2) = 2, \quad r_2 = \max(5, 2) = 5$$

Thus, the queried subarray is $a_{2...5} = [3, 2, 3, 4]$. Values 2 and 4 appear once (odd), and value 3 appears twice (even). Hence, the beauty is $2 + 4 = 6$.

In the third test case, all elements of array a are equal to 3.

- For any subarray of odd length, the value 3 appears an odd number of times, so the beauty is 3.
- For any subarray of even length, the value 3 appears an even number of times, so the beauty is 0.



Server time: Oct/10/2025 22:10:04^{UTC+7} (f1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



ITMO